

Curriculum Vitae

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Education

2025: PhD in chemistry (ENS de Lyon, France).
2021: Master's degree in chemistry in computational chemistry (ENS de Lyon, France).
2019: Degree in Chemistry and Physics with *Mention TB* (ENS de Lyon).
2016-2018: "Classes préparatoires" in maths, physics and chemistry, top of class (Chateaubriand, Rennes, France)
2016: Scientific Baccalaureate, average of 18.44/20 (Assomption, Rennes, France)

Work Experience

2025 – present: post-doc at KU Leuven with Prof. Jeremy Harvey, working on various topics in catalysis and computational chemistry.

2022 – 2025 present: PhD student at ENS de Lyon with Dr Carine Michel, working on the chemistry of ruthenium clusters on TiO₂. Teaching tutorials in solid state computational chemistry and statistical physics.

March – July 2022: internship at Université Grenoble-Alpes with Prof. Anne Milet on the computational study of catalysts for reducing CO₂ (CP2K, DockOnSurf).

November 2021 – February 2021: internship at the University of Durham with Dr Basile Curchod, working on the limits of TD-DFT for studying atmospheric molecules (Newton-X, Turbomole).

February – July 2021: 6-month internship in the chemistry laboratory of the ENS de Lyon with Dr Stephan STEINMANN working on resampling techniques for calculating adsorption free energies (SolvHybrid, AMBER, VASP and Python3)

May - July 2020: 3-month internship in the chemistry laboratory of the ENS de Lyon with Dr Carine MICHEL investigating the adsorption of alcohols on alumina using metadynamics (CP2K and PLUMMED)

January - April 2020: part time internship in chemistry laboratory of the ENS de Lyon with Dr Carine MICHEL working on the quenching of metathesis catalysts in solvents containing traces of water (Python, ANAKIN-ME)

April 2020: co-author of a problem for the French selection competition for the 52nd International Chemistry Olympiad (<https://www.sciencesalecole.org/wp-content/uploads/2021/01/sujet-ICHO-2020.pdf>, exercise Zimmerman 1)

July 2019: Volunteer work as a scientific administrator at the 51st International Chemistry Olympiad in Paris

June - July 2019: 2-month internship at the University of Rennes 1 with Pr. Florence MONGIN and Dr. William ERB working on developing new methods for enantioselective synthesis of ferrocenes.

Publications

Sims, J. M.; Michel, C. ; "Influence of oxygen vacancies on the morphology of Ru₁₀ clusters on supported TiO₂", *Physical Chemistry Chemical Physics*, vol. 27, pp. 17564-17571. <https://doi.org/10.1039/D5CP01579K>

Utama, J. K.; Sims, J. M.; Michel, C. ; Giorgi, J. B.; "Hydration of Hexafluoroacetone on Ice", *Journal of Physical Chemistry C*, vol. 129, no. 12, pp. 5868–5875, 2025. <https://doi.org/10.1021/acs.jpcc.5c00560>

Berg, I.; Mondal, R.; Sims, J. M.; Ben-Tzvi, T.; Friedman, B.; Michel, C.; Nairoukh, Z.; Gross, E.; "Strong Substrate-Adsorbate Interactions Direct the Impact of Fluorinated N-Heterocyclic Carbene Monolayers on Au Surface Properties." *ACS Applied Materials and Interfaces*, vol. 16, no. 47, 2024, pp. 65469-65479. <https://doi.org/10.1021/acsami.4c12514>

Arjunan, S.; Sims, J. M.; Duboc, C.; Maldivi, P.; Milet, A.; "Investigating the interplay between charge transfer and CO₂ insertion in the adsorption of a NiFe catalyst for CO₂ electroreduction on a graphite support through DFT computational

approaches." *Journal of Computational Chemistry*, vol. 45, issue 19, 2024, pp. 1690-1696.

<https://doi.org/10.1002/jcc.27355>

Blanco, C. O. ; Sims, J. ; Nascimento, D. L.; Goudreault, A. Y.; Steinmann, S. N.; Michel, C.; Fogg, D. E.; "The Impact of Water on Ru-Catalyzed Olefin Metathesis: Potent Deactivating Effects Even at Low Water Concentrations." *ACS Catalysis*, vol. 11, no. 2, Jan. 2021, pp. 893–99, <https://doi.org/10.1021/acscatal.0c04279>.

Lassagne, F. ; Sims, J. M.; Erb, W.; Mongin, O.; Richy, N.; El Osmani, N.; Fajloun, Z.; Picot, L.; Thiéry, V.; Robert, T.; Bach, S.; Dorcet, V.; Roisnel, T.; Mongin, F.; "Thiazolo[5,4-f]Quinoxalines, Oxazolo[5,4-f]Quinoxalines and Pyrazino[b,e]Isatins: Synthesis from 6-Aminoquinoxalines and Properties." *European Journal of Organic Chemistry*, vol. 2021, no. 19, 2021, pp. 2756–63. <https://doi.org/10.1002/ejoc.202100362>.

Reviewer: JACS

Conferences

Sims, J. M.; Michel, C., "Morphology of Ru nanoparticles at the titania-water interface – A computational study", oral presentation, International Congress on Theoretical Aspects of Catalysis.

Sims, J. M.; Michel, C., "Morphology of Ru nanoparticles at the titania-water interface – A computational study", poster, International Congress on Catalysis 2024.

Sims, J. M.; Michel, C., "Morphology of Ru nanoparticles at the titania-water interface – A computational study", oral presentation, Rencontres Théomsia-RCTF 2024 (conference of the French Computational Chemistry Network).

Milet, A.; Maldivi P.; Arjunan S.; Sims J. M., "Theoretical Mechanistical Study of CO₂ Reduction to CH₄ by a Bio-Inspired NiFe Hydrogenase Model on graphite", poster, 16th European Inorganic Chemistry Conference, 2022.

Languages

English (native speaker, Cambridge Advanced Certificate C2)

French (bilingual)

German (B1)